

Pierre Antoine Delice

AI Engineer & NLP Researcher | Data Scientist

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Professional Profile

AI engineer, NLP researcher, and data scientist with a Ph.D. in Language and Knowledge Engineering and more than 15 years of experience applying statistical modeling, machine learning, administrative data, and evidence systems to public, economic, health, and education-sector decisions. Recent work focuses on LLM evaluation and prompting, conversational AI, Spanish-language NLP, transformer-based domain adaptation, economic news analytics, information retrieval, and record linkage. Comfortable moving between research prototypes, production-oriented data products, teaching, and senior stakeholder advisory roles.

Technical Skills

Programming	Python, R
AI / NLP	Large language models, prompt engineering, LLM evaluation, natural language processing, transformer-based domain adaptation, sentiment analysis, information retrieval, argument mining, pattern recognition, deep learning, fuzzy models
Data Science	Statistical modeling, survey design, record linkage, administrative and vital-record data, unstructured-data analysis, economic indicator inference, costing analysis, dashboards, geospatial visualization, monitoring and evaluation
Domains	Conversational AI for education, economic news analytics, public health data, violence and security analytics, poverty measurement, applied policy analysis

Professional Experience

AI Engineer <i>Laureate International Universities via Northware</i>	Feb 2025 – Present <i>Mexico</i>
- Developed a conversational AI system for students and prospects across UNITEC/UVM. - Built and improved a WhatsApp registration and enrollment workflow, including payment support, so the chatbot can operate closer to a human enrollment adviser. - Analyze and adapt LLM behavior for chatbot use cases through prompt design, instruction testing, and qualitative response evaluation.	
Adjunct Professor / Profesor Catedra <i>Universidad de Guadalajara</i>	Aug 2025 – Present <i>Jalisco, Mexico</i>
- Teach undergraduate courses in applied programming, Programming II, applied mathematics, and sustainable projects. - Design applied learning materials connecting programming, data analysis, mathematics, and engineering practice.	
Data Analyst <i>National Council of Science and Technology (CONAHCYT)</i>	Apr 2023 – Dec 2024 <i>Mexico</i>
- Designed and coordinated the National Survey for Mental Health and Addictions (ENASAMA 2023–2024). - Supported statistical design, data organization, and evidence generation for national research and policy analysis.	
Data Analyst / Costing Consultant <i>Catholic Relief Services (CRS)</i>	Jun 2023 – Jan 2024 <i>Mexico</i>
- Conducted costing analysis for the Grupo de Apoyo a Mujeres program using unstructured datasets from multiple sources. - Produced a costing exercise to support program planning, monitoring, and resource allocation decisions.	
Technical Advisor and Data Consultant <i>PAHO, Guerrero Ministry of Health, UNODC, CONEVAL</i>	2016 – 2023 <i>Mexico</i>
- Developed monitoring and evaluation work for community COVID-19 mitigation with the Pan American Health Organization. - Produced diagnostics of violence in Guerrero using survey data and structured interviews, and contributed to public security strategy design. - Designed and implemented the Forensic Medicine Specialty program in Guerrero, including academic standards,	

faculty recruitment, and accreditation follow-up.

- Designed a proposal for a state-level monitoring and evaluation system for CONEVAL.

Senior Data and Policy Roles

2008 – 2016

Health, economic development, HIV/AIDS, and planning institutions

Mexico / Haiti

- Managed and analyzed national vital statistics datasets, including births, deaths, and causes of death, for epidemiological surveillance and public policy.
- Built dashboards, data models, and statistical analyses for hospital performance, service coverage, health indicators, and resource allocation.
- Developed composite indexes and geospatial visualizations for Mexico City competitiveness and urban policy analysis.
- Produced HIV/AIDS prevalence and incidence estimates and projections for national and regional strategic planning.
- Conducted cost analysis for health and agricultural sectors for Haiti's Poverty Reduction Strategy Paper.

Selected AI, NLP, and Data Work

- **Economic news NLP:** Built research on Spanish-language economic news as signals for economic analysis and inflation-related dynamics, published in *Journal of Scientometric Research* and *IEEE LAEDC*.
- **Conversational AI:** Developed education-sector chatbot workflows for student advising, registration, and payments over WhatsApp.
- **LLM ranking:** Authored 2026 work on listwise LLM ranking for multimodal Spanish political headline classification.
- **Domain-adapted sentiment analysis:** Co-authored transformer-based domain adaptation work for Mexican tourism reviews at IberLEF 2023.
- **Record linkage and administrative data:** Worked on birth/death linkage, similarity metrics, vital statistics, and health-sector monitoring datasets.

Education

Ph.D., Language and Knowledge Engineering / Computer Science , Benemerita Universidad Autonoma de Puebla (BUAP)	2025
M.Sc., Government and Public Affairs , Facultad Latinoamericana de Ciencias Sociales (FLACSO), Mexico	2010
B.A., Economics, Planning Specialization , Centre de Techniques de Planification et d'Economie Appliquee (CTPEA), Haiti	2006

Selected Publications

- Delice, P. A. & Pinto, D. (2025). "Decoding Economic Insights: The Analytical Power of News Content." *Journal of Scientometric Research*, 14(1), 365–372. DOI: [10.5530/jscires.20251459](https://doi.org/10.5530/jscires.20251459).
- P. A. Delice, D. Pinto, V. M. Garcia-Guerrero, and S. Hernandez-Lopez (2024). "The Economic Value of Words: An Evaluation of News for Economic Analysis." *IEEE Latin American Electron Devices Conference (LAEDC)*. DOI: [10.1109/LAEDC61552.2024.10555884](https://doi.org/10.1109/LAEDC61552.2024.10555884).
- Tellez-Velazquez, A., Delice, P. A., Salgado-Leyva, R., and Cruz-Barbosa, R. (2024). "On the Explanation of COVID-19 Blood Test Variables Using Fuzzy Models." *Journal of Intelligent & Fuzzy Systems*. DOI: [10.3233/JIFS-219372](https://doi.org/10.3233/JIFS-219372).
- Morales-Murillo, V. G., Gomez-Adorno, H., Pinto, D., Cortes-Miranda, I. A., and Delice, P. (2023). "LKE-IIMAS Team at Rest-Mex 2023: Sentiment Analysis on Mexican Tourism Reviews Using Transformer-Based Domain Adaptation." *IberLEF 2023*.
- Delice, P. A. (2026). "LKE-BUAP at PoliticHeadlines-IberLEF 2026: Listwise LLM Ranking for Multimodal Spanish Political Headline Classification." *CEUR Workshop Proceedings*.
- Delice, P. A. (2026). "IA y empoderamiento economico: lecciones desde proyectos comunitarios." Accepted book chapter, UPAEP.

Selected Presentations, Honors, and Service

- IEEE Senior Member, 2025.

- Publications Chair, IEEE Latin American Electron Devices Conference (LAEDC), 2025.
- Presented economic news NLP research at LKE 2024, IEEE LAEDC 2024, CIBERTIC 2025, and IEEE IAS Electrical Safety Workshop 2025.
- CLACSO Research Grant, 2021; CLACSO Young Researchers Grant, 2012.
- Mexico–Chile Bilateral Fund / AMEXCID award for forensic-science training and institutional cooperation, 2020.
- Evaluator for Vive la Ciencia 2024 technical review committee, CONCYTEP.

Training and Languages

Training	Deep Learning Course, Neuromatch Academy, 128 hours, 2023; Multidimensional Poverty Measurement and Applications, ECLAC / OPHI / MIDEPLAN, 2010
Languages	Spanish, English, French